Petrol Pump Management System

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Course:
Object Oriented Programming

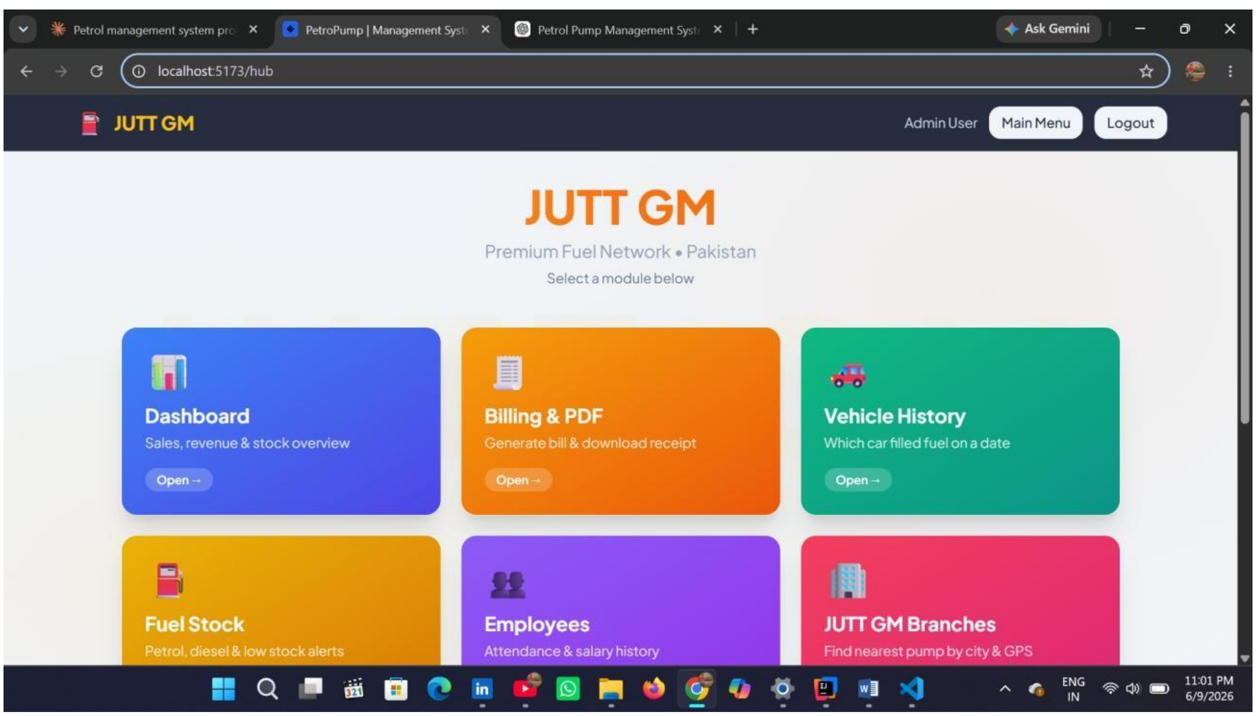
Department:
Computer Engineering

Abstract:

The Petrol Pump Management System is a full-stack web application developed to automate daily operations of a petrol station. The system manages fuel inventory, billing, employee records, attendance, salary management, reporting, and user authentication.

The application follows the MVC (Model View Controller) architecture and applies Object-Oriented Programming principles to ensure scalability, maintainability, and code reusability.

The frontend is developed using React and Vite while the backend uses Node.js, Express.js, Sequelize ORM, and MySQL Database.



Introduction

Managing a petrol pump manually can lead to:

- Human errors
- Billing mistakes
- Fuel stock mismanagement
- Employee record issues
- Delayed report generation

The Petrol Pump Management System provides a digital solution that automates these operations and improves efficiency.

Objectives

- Manage fuel inventory
- Generate customer bills
- Maintain employee records
- Generate reports
- Secure user authentication
- Improve operational efficiency

Problem Statement

Traditional petrol stations rely heavily on manual record keeping.

Challenges include:

- Inaccurate fuel stock calculations
- Time-consuming billing

- Difficult employee management
- Lack of real-time reports
- Data redundancy

The proposed system addresses these challenges through automation and centralized data management.

MVC Architecture

Model

Responsible for:

- Database interaction
- Business entities
- Data validation Examples:

- User Model
- Fuel Model
- Employee Model
- Billing Model

View

Responsible for:

- User Interface
- Dashboard
- Forms
- Reports

Implemented using:

- React.js
- Tailwind CSS

Controller

Responsible for:

- Handling requests
- Processing business logic

- Returning responses Implemented using:

- Express Controllers

Diagram

```
User
 |
 V
View (React)
 |
 V
Controller (Express)
 |
 V
Model (Sequelize)
 |
 V
MySQL Database
```

Technology Stack

Technology Stack

Frontend

- React 18
- Vite
- Tailwind CSS
- Axios
- Chart.js **Backend**

- Node.js
- Express.js
- JWT Authentication
- bcrypt
- Multer
- Swagger

Database

- MySQL

ORM

- Sequelize

Development Tools

- VS Code
- Git
- Postman

OOP Concepts Used

Object-Oriented Programming

The project extensively uses OOP concepts.

Class

A blueprint for creating objects.

Examples:

- User
- Employee
- Fuel
- Billing

Object

An instance of a class.

Example:

```
Employee emp = new Employee();
```

Benefits

- Better code organization
- Reusability
- Easy maintenance

Four Pillars of OOP

1. Encapsulation

Bundling data and methods together.

Example:

```
private String fuelName;

public String getFuelName()
{     return
fuelName;
}
```

2. Inheritance

Allows child classes to acquire parent properties.

Example:

```
class User
class Admin extends User
```

3. Polymorphism

Single interface with multiple implementations.

Example:

```
generateReport()
```

can generate:

- Daily Report
- Monthly Report
- Revenue Report

4. Abstraction

Showing essential details while hiding implementation.

Example:

API endpoints expose functionality without revealing internal logic.

Database Design

Database Modules

Users Table

- id
- name
- email
- password
- role

Fuel Table

- id
- fuel_name
- quantity
- price

Employees Table

- employee_id
- employee_name
- salary

Billing Table

- bill_id
- customer_name
- amount

Reports Table

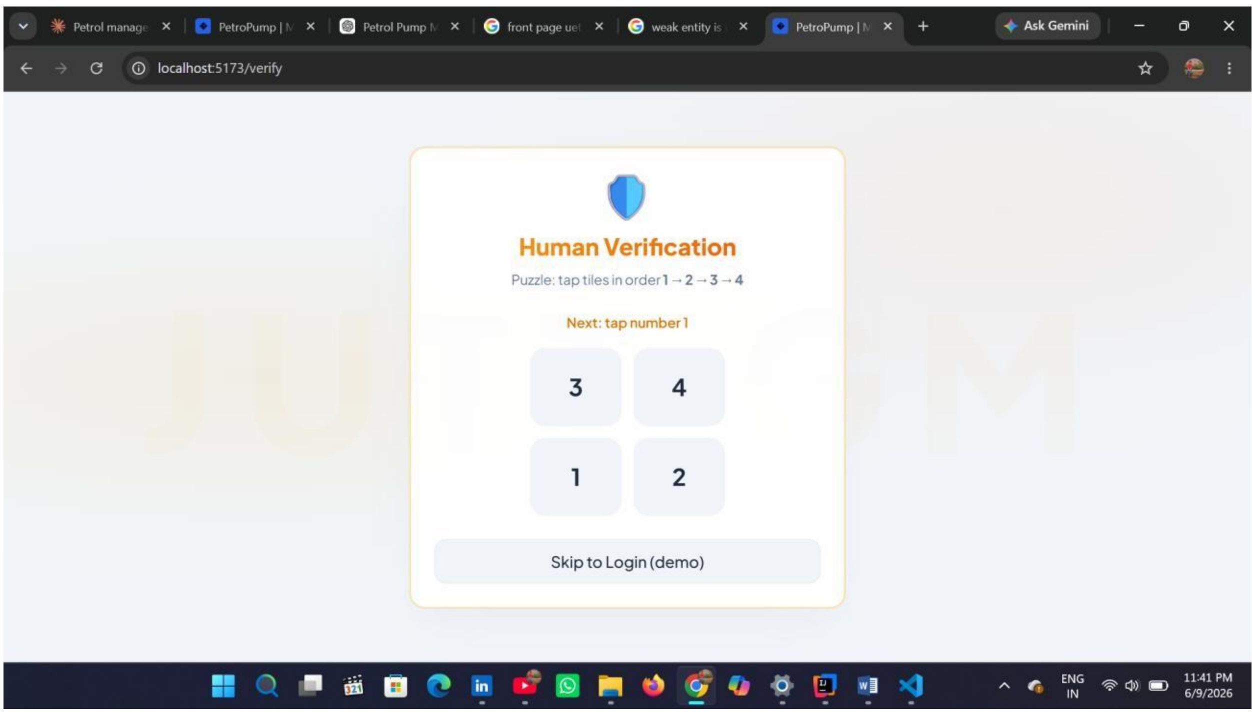
- report_id

System Features

Main Features

Authentication

- JWT Login
- Role Based Access



Fuel Management

- Add Fuel
- Update Fuel
- Stock Monitoring

Billing

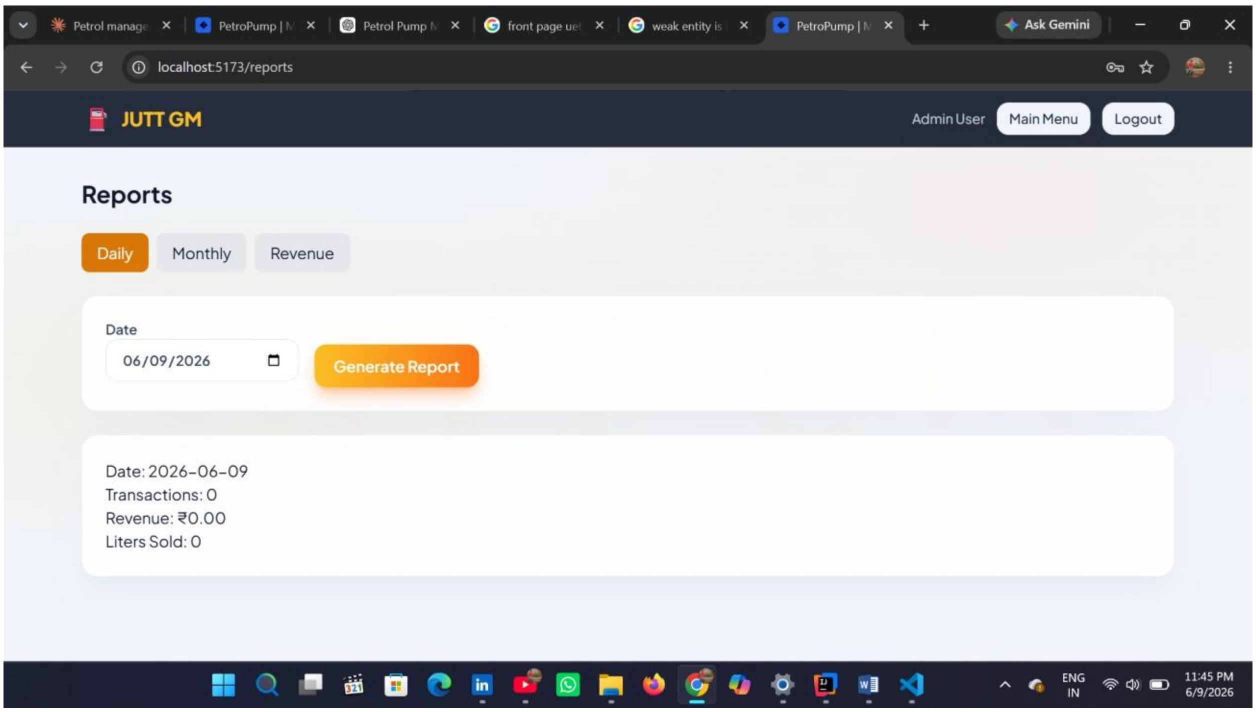
- GST Calculation
- Receipt Printing

Employee Management

- Attendance
- Salary Records

Reports

- Daily Reports
- Monthly Reports
- Revenue Reports



Important Modules

Dashboard Module

Provides:

- Fuel Analytics
- Revenue Summary

Inventory Module

Provides:

- Fuel Tracking

- Low Stock Alerts

Employee Module

Provides:

- Employee Records
- Salary Information

Billing Module

Provides:

- Bill Generation
- Printable Receipts

Advantages of MVC

Advantages of MVC Architecture

Separation of Concerns

Each component has a specific responsibility.

Easy Maintenance

Changes in UI do not affect database code.

Scalability

New modules can be added easily.

Reusability

Components can be reused across the application.

Better Team Collaboration

Frontend and Backend teams can work independently.

Real Life Importance

Importance in Daily Life

The system helps petrol stations by:

- Reducing manual work
- Improving accuracy
- Saving time
- Managing employees efficiently
- Generating instant reports

Business Benefits

- Better decision making
- Improved customer service
- Reduced operational costs
- Increased productivity

Resume and Job Market Importance

Importance for Resume

This project demonstrates skills in:

Programming

- Java Concepts
- JavaScript
- OOP

Web Development

- React
- Node.js
- Express

Database

- MySQL

- Sequelize ORM

Software Engineering

- MVC Architecture
- Authentication
- API Development **Recruiter**

Perspective

This project showcases:

- Full Stack Development
- Problem Solving
- Database Management
- Software Design Skills

These are highly demanded skills in software development and cybersecurity industries.

Conclusion and Future Enhancements

Conclusion

The Petrol Pump Management System successfully automates major operations of a petrol station.

The system improves:

- Efficiency
- Accuracy
- Security
- Productivity

By implementing MVC Architecture and Object-Oriented Programming principles, the project achieves a clean, scalable, and maintainable software structure.

Future Enhancements

- AI-based fuel demand prediction
- Online payments
- SMS notifications
- Mobile application
- Cloud deployment
- Multi-branch support